

CS 6501 – Week 6 Cheat Sheet + Final Project Notes

1. Social Compliance in Robotics

Definition: Social compliance goes beyond:

- Safety (no collisions)
- Liveness (eventual task completion)

It requires behavior that aligns with human expectations, norms, and contextual appropriateness.

Observation Function

Robot-Robot:

$$O_t^i = \{x_t^i\} \cup \{x_t^j, j \in \mathcal{N}(x_t^i)\}$$

Robot-Human:

$$O_t^i = O_t^i \cup \{RGB, LiDAR, Depth\}$$

Pipeline:

$$x_t^i \rightarrow o_t^i \rightarrow \text{Perception/Reasoning} \rightarrow \pi_t^i \rightarrow u_t^i$$

Key insight: Social compliance depends heavily on perception and reasoning layers, not just control.

2. VLM-Social-Nav

Formulated as an MDP:

$$(S, A, T, C)$$

Cost function:

$$C(s, a) = \alpha C_{goal} + \beta C_{obst} + \gamma C_{social}$$

Social cost:

$$C_{social} = \|B - B_h\|$$

Where:

- B = robot behavior
- B_h = human-like socially compliant behavior

VLM inference:

$$B_h^{t+1} = \text{VLM}(I_t, P, a_t)$$

Updated social cost:

$$C_{social}^{t+1} = w_l|v - v_h| + w_a|\omega - \omega_h|$$

Architecture:

- Perception (YOLO detects humans/doors/gestures)
- VLM scoring module computes social cost
- Low-level planner (e.g., DWA) minimizes total cost

Limitation:

- VLM inference introduces latency (seconds scale)
 - Requires gating mechanism for real-time deployment
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3. Learning Social Behavior from Dataset

Alternative approach: imitation / IRL.

$$J_{social} = \left\| \mathbb{E}_{\Gamma_\theta \sim D_\theta} \Phi(\Gamma_\theta) - \mathbb{E}_{\hat{\Gamma} \sim \hat{D}} \Phi(\hat{\Gamma}) \right\|$$

Pipeline:

$$x_t^i \rightarrow o_t^i \rightarrow \text{Encoder} \rightarrow u_t^i$$

Limitations:

- Requires large human datasets
 - Poor real-world generalization
 - Training instability
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4. Costmap Layering

Idea:

- Modify costmap instead of controller
- Regions frequently used by humans assigned higher cost
- A* planner avoids socially inappropriate areas

Pros:

- Stable
- No neural network
- Engineering-friendly

Cons:

- Not generalizable
 - Rigid
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5. GAMEOPT – Hybrid Auction + Optimization

Two-phase approach:

1. Auction selects priority sequence
2. Optimization computes safe trajectories

Objective:

$$\min_{\{u_i\}} \max_i t_i^f$$

Priority value:

$$\zeta_i = s_i(t) \cdot (c - \tau_i)$$

Guarantees:

- Safety
- Fairness (incentive compatibility)
- Efficiency
- Real-time computation

Key idea: Separate combinatorial ordering from trajectory optimization.

6. Trajectory Prediction Context

General formulation:

Historical trajectory:

$$x_{t-H:t}^{1:k}$$

Future prediction:

$$x_{t+1:t+T}^{1:k}$$

Observation model:

$$O_t^i = \{x_t^i\} \cup \{x_t^j, j \in \mathcal{N}(x_t^i)\} \cup M_t^i$$

Limitations:

- Strong in benchmarks
 - Weak real-world transfer
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Final Project: VIP-Constrained Multi-Agent Coordination

Core Question

How can we incorporate strict agent priority into MAPF so that a designated VIP agent maintains trajectory optimally while other agents adapt through asymmetric conflict resolution?

Motivation

- Multi-drone systems in surveillance, escort, disaster response
- Many deployments require hierarchical structure
- Standard MAPF assumes equal priority
- Real systems require asymmetric conflict resolution

Research Axes

1. Greedy vs Long-Term Optimality

- Does greedy priority create bottlenecks?
- Can short-term sacrifice improve long-term flow?
- SMG doorway scenarios
- Roundabout / merging known-map reasoning

Central question:

Is greedy priority globally optimal?

2. Ambiguous Priority Scenarios

Examples:

- Routine hospital visit vs late flight
- Routine vs emergency

Questions:

- How to encode contextual priority?
- Static vs dynamic priority?
- LLM reasoning layer?

3. VIP Formation Variation

Scenario:

- VIP shortest path
- Followers longer path
- VIP waits to regroup

Tradeoff:

- Individual autonomy
- Group-level objective

4. n-SMG Perspective

- Structured mini-games
- Bottleneck modeling
- Compare local vs global reward

Is there a better strategy than greedy priority assignment?

5. LLM Integration

Issues:

- LLM latency
- Dialogue during navigation
- Near real-time reasoning

Demo concept:

- Standard SMGLib
- LLM-enhanced version
- Toggle reasoning

6. Research vs Engineering Direction

Research-focused:

- Asymmetric CBS conflict trees
- Priority enforcement analysis
- Theoretical guarantees

Engineering-focused:

- Real-time IMPC + LLM
- Latency gating
- Production-ready demo